The Contact Process

Exam code: 0620 & 0971

Download PDF

Written by: Alexandra Brennan
Reviewed by: Stewart Hird
Updated on 25 November 2024

SaveMyExams

Did this video help you? Yes No

The Contact process

Extended tier only

- Sulfuric acid is synthesised by the **Contact process**
- Concentrated sulfuric acid is used in car batteries, making **fertilisers**, **soaps** and **detergents**

Stage 1

- The first stage is the production of sulfur dioxide
- The oxygen for this stage is obtained from the air
- The sulfur for this stage is obtained by:
  - Burning sulfur to oxidise the sulfur

sulfur + oxygen → sulfur dioxide

$$S + O_2 \rightarrow SO_2$$

- Roasting sulfide ores

metal sulfide + oxygen → metal oxide + sulfur dioxide

Stage 2

- The main stage is the oxidation of sulfur dioxide to sulfur trioxide using a **vanadium(V) oxide, V_2O_5, catalyst**

sulfur dioxide + oxygen ⇌ sulfur trioxide

$$2SO_2 + O_2 \rightleftharpoons 2SO_3$$

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- A pressure of **2 atm (200 kPa)**

- Once sulfur trioxide is formed, it undergoes more processes to produce sulfuric acid

Stage 3

- The sulfur trioxide is absorbed into a solution of 98% sulphuric acid to produce a thick liquid called **oleum (**disulfuric acid):

$$SO_3 + H_2SO_4 \rightarrow H_2S_2O_7$$

sulfur trioxide + sulfuric acid → disulfuric acid

- It is not absorbed into water because a fine mist of sulfuric acid would be produced and this would be difficult to condense and is also highly dangerous
- Oleum is added to water to form concentrated sulfuric acid:

$$H_2O + H_2S_2O_7 \rightarrow 2H_2SO_4$$

water + disulfuric acid → sulfuric acid

Examiner Tips and Tricks

You need to recall the temperature, pressure and catalyst needed for the Contact process and the equation for stage 2 only.

Explaining the conditions in the Contact process

Extended tier only

- Similar to the Haber process, the pressure and temperature used need to be considered
- The equation for the main stage of the Contact process is:

$$2SO_2 + O_2 \rightleftharpoons 2SO_3$$

Temperature: 450 °C

- **High** temperature favours the reverse reaction as it is endothermic
  - So, a higher yield of **reactants** will be made

- **Low** temperature favours the forward reaction as it is exothermic
  - So, a higher yield of **products** will be made

- However, at low temperature the rate of reaction is very **slow**
- So, 450 °C is a compromise temperature between having a **lower yield** of products but being made more quickly

more moles of gaseous reactant
    - So, a higher yield of **reactants** will be made
- **High** pressure favours the forward reaction as there are **fewer** moles of gaseous product
    - So, a higher yield of **products** will be made
- However, the position of equilibrium lies far to the right
    - The equilibrium mixture contains about 96% sulfur trioxide
- So, the reaction is carried out at just above atmospheric pressure because:
    - High pressures can be dangerous and very expensive equipment is needed
    - A higher pressure causes the sulfur dioxide to liquify

Examiner Tips and Tricks

Remember: These conditions are a **compromise** between yield, rate, safety and cost.

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Explore in Test Builder →

Test yourself

Flashcards

Was this revision note helpful? Yes No

Resources

Learning Hub

Scholarship

Join

Past Papers

Solution Banks

Sitemap

Dashboard

Overview

Brand ambassador

Profile

Subscription

Settings

Log out

Company

About us

Exam Specificity

Content Quality

Promotions

Jobs

Terms

Privacy

Cookie Policy

Help and Support

Subjects

Biology

Chemistry

Physics

Maths

Geography

English Literature

Psychology

All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.

Neither IBO nor College Board were involved in the production of, and do not endorse, the resources created by Save My Exams.